Nested-quotient temporal histories and homological flat bands in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Final edition, with machine-checked provenance

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. After selecting the complete gauge-invariant charge-odd one-plaquette eigenspace, fixed-order contributions are organised as ordered magnetic histories on the Haar–Gram quotient of reachable Wilson networks. The complete momentum dependence through second order is the oriented edge–face Gram operator $t_N u^2 B(k)B(k)^\dagger$ with $B = \partial_2^\dagger$ and

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0, \qquad N \geq 3.$$

The projected spectrum is $\{0,\, t_N u^2 q,\, t_N u^2 q\}$ up to a common scalar, with $q = 4\sum_j \sin^2(k_j/2)$; the flat carrier is $\ker \partial_2$, of exact dimension $L^3 + 2$.

Relative to the 28 August edition this one adds five things, each carrying its own machine check. The channel decomposition of t_N, previously a representation-theoretic ledger derived on a single plaquette pair, is recovered on a genuine face-sharing two-cube $SU(3)$ space of 1,590,462 states, and the two decompositions agree *channel by channel* rather than only in the sum (Reported result 12); on that space each of the six channels is separately proportional to the same incidence Gram, on all 56 adjacent pairs, so the separation of colour from geometry is measured and not assumed, and the connected diagonal — eleven numbers in both deliveries — is constant on the prism's three symmetry orbits, with the whole $B = 4 \to B = 6$ difference confined to the eight faces that carry connected transport (Proposition 16). Separately, and by arithmetic on the weight ledger alone, a cutoff keeping only the singlet and $\Lambda^2 F$ routes projects the hopping to $-1/12$ — opposite in sign to t_3 — and omits exactly $14/153$, the two summing to t_3 (Corollary 14); the $B = 4$ truncation of the two-cube space delivers the first of those two numbers outright, on the same geometry. The charge-even sector acquires a closed form: with $\hat{a}_m = 2 + 2\cos k_m$ and $p = \sum_m \hat{a}_m = 12 - q$, its Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$, so one zone function runs both sectors (Theorem 19), and its Γ-point splitting pins the charge-even hopping $t_{r,+}$ — a separation the charge-odd rest frame has no analogue for, since its Γ spectrum has only one level (Proposition 22). The homological reading of the flat direction is shown to be the same one on a second complex: the six faces of a cube are a closed surface, its $\ker \partial_2$ is the fundamental class, and the one-cube charge-odd shell splits as $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$ with the flat level sitting at the scalar for *every* value of the hopping (Proposition 17). Finally the open fourth-order off-axis coefficient is given a geography rather than a verdict: an exact obstruction certificate, an explicit second witness, and the measurement that no Γ-anchoring error can move it at all (Section 9).

$SU(3)$ third- and fourth-order statements remain conditional on supplied exact-arithmetic ledgers whose raw payloads are not reproduced here, and are labelled as such throughout. Every displayed result carries the name of the machine check that establishes it. All statements concern a formal finite-order projected sector: this is not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [1, 2]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology, line-graph constructions, compact localised states, and the need for noncontractible states on periodic systems are all established, and closely related closed-surface structures occur in two-form spin liquids. A recent classification of exactly flat bands protected by local graph topology [14] reaches extensive face-graph degeneracy from an *unoriented* face–vertex incidence, where rank–nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face–edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal L^3+2 — homology, not counting. It is recorded as a distinction, not an identification. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

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check: the discrete index theorem gives 0 on the signed
face-edge operator, bounding nothing
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The content is dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an oriented edge–face Gram operator in the charge-odd one-plaquette sector. Representation theory fixes its coefficient and the cubical chain complex fixes its kernel and spectrum. The separation is exact,

$$\text{colour dynamics} \longrightarrow t_N,$$
$$\text{cellular geometry} \longrightarrow BB^\dagger.$$

What is proved, and what is conditional

Seven statements are self-contained here.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2, derived from the order-two Weingarten values (Theorem 6). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.

2. For every $N \geq 3$ the complete momentum dependence at order u^2 is $t_N u^2 BB^\dagger$ (Theorem 9).

3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 1), and on the periodic L^3 complex the zero-mode count is L^3+2 (Theorem 2). The two are the same statement, by a sum rule over the momentum grid (Theorem 4).

4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_{\mathbf{1}} = -1/12$, and what it omits is $w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = 14/153$; the two sum to t_3 (Corollary 14). This is arithmetic on (11) and needs no input beyond Theorem 6.

5. The charge-even Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 19); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 20); and its Γ-point splitting is $12t_{r,+}$ exactly (Proposition 22). The identity is self-contained; the two numbers it is evaluated at inherit the status of $t_{r,+}$, and the $r = 3$ one is conditional — see item 9.

6. On the closed cube surface the charge-odd shell is $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$, with G-eigenvalues $0, 4, 6$; the labels follow from the rotation action rather than being assigned, the A_1^{--} level is the fundamental class of S^2, and it sits at the scalar for every hopping coefficient (Proposition 17). This is integer linear algebra and takes no input from any delivery.

7. The retained Γ/axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 27), and an explicit second witness exhibits the non-identifiability by construction (Proposition 29).

Two families of statements are *conditional* on supplied exact-arithmetic derivation ledgers. Their raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not included here, and repeated assertions in synthesis documents are not treated as independent reproductions.

8. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 12) — and the $B = 4$ truncation of that space is the same truncation as the published link cutoff of item 4. Conditional on the delivered diagonals, the symmetry analysis of Proposition 16 applies to them: the orbit structure itself is exact and takes no delivered input. The 1,590,462-state build was not reproduced here.

9. For $SU(3)$, supplied ledgers report that all candidate third-order lifter classes vanish and report the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, and nothing here checks any of them. If it is exhaustive and the linked ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3.

10. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2-1)^3]$ and its equality with the complete axial coefficient for $N = 3, \ldots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4. The

complete off-axis operator remains open — and by Theorem 27 it cannot be closed by any Γ or axial datum.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2}. Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2}\sum_\ell C_2(\ell) + \beta_N \sum_p \Big(1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p\Big). \quad (1)$$

Dropping the additive constant,

$$H = H_E - uV, \qquad V = \sum_p (\chi_p + \bar\chi_p), \qquad u = \frac{\beta_N}{2N}, \quad (2)$$

with $\chi_p = \operatorname{Tr} U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$-specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u. Reading it as a coordinate would demand dividing the order-r coefficient by 4^r, which misses the printed tables by exactly 16 at u^2 and 64 at u^3.

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check: the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim
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check: the 4**r rescaling breaks the bridge: order 2 off by 16, order 3 by 64
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All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\mathrm{exc}}(u) = H(u) - E_{\mathrm{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar\chi_p}{\sqrt{2}}\,|0\rangle, \quad (3)$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\Big(\frac{1}{2}C_F\Big) = 2C_F = \frac{N^2-1}{N}, \qquad C_F = \frac{N^2-1}{2N}. \quad (4)$$

This external energy is the input to the resolvent denominators of Section 4, where it is checked together with them. For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The charge-even hopping does *not* vanish there, which locates the $N^2 - 4$ zero in the channel difference rather than in the incidence.

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check: at N = 2 the C-odd hopping vanishes and the C-even one does not
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We work inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric centre flux around each torus cycle. Let P_- be the full $E_{F,N}$ spectral projector there. For $L \geq 3$ its range is the charge-odd one-plaquette span, since a reduced contractible fundamental loop at that energy has four distinct links and is an elementary plaquette. This sector qualification also excludes the accidentally degenerate straight wrapping loop at $L = 4$. With $Q = 1_- - P_-$ the reduced resolvent crosses no omitted $E_{F,N}$ eigendirection, and degenerate perturbation theory gives

$$H_{\mathrm{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \cdots \quad (5)$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q-space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2\partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1\partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4\sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (6)$$

Theorem 1 (Incidence factorisation)**.** *With* $w(k) = (\bar d_3, -\bar d_2, \bar d_1)^\top$,

$$BB^\dagger = qI - ww^\dagger, \qquad B^\dagger w = 0, \qquad \|w\|^2 = q,$$

and therefore $\operatorname{spec}(BB^\dagger) = \{0, q, q\}$. *With* $S = BB^\dagger - 4I$ *the signed shared-link adjacency,* $\operatorname{spec} S(k) = \{-4, q-4, q-4\}$.

Proof. Multiplying (6) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. □

`check: B B^dagger = q I - d conj(d)^T for the curl incidence`

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ, $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 2 (Finite-torus carrier). *For the periodic cubic cellulation of* T^3, $\ker \partial_2 = \operatorname{im} \partial_3 \oplus H_2$ *with* $\dim \operatorname{im} \partial_3 = L^3 - 1$ *and* $\dim H_2 = 3$, *so* $\dim \ker \partial_2 = L^3 + 2$.

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\operatorname{rank} \partial_3 = L^3 - 1$ by rank–nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. □

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \ldots, 5$.

`check: dim Z_2 = L^3 + 2 by rank, not by re-arranging the formula`

`check: cube boundaries and three wrapping sheets SPAN Z_2`

Remark 3. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x+e$ and $x-e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

`check: the L^3+2 count is chain-level, not the Bloch convention`

The following sum rule joins Theorems 1 and 2, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$-dimensional space.

Theorem 4 (Bloch–chain bridge). *On the periodic* L^3 *complex,*

$$\sum_k \bigl(3 - \operatorname{rank} B(k)\bigr) = L^3 + 2,$$

the sum running over the L^3 *allowed momenta, with nullity profile* 3 *at* Γ *exactly once and* 1 *at each of the other* $L^3 - 1$ *momenta.*

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 1 gives $\operatorname{rank} B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. □

`check: the Bloch and chain routes to the carrier agree`

Remark 5. The two decompositions produce the same two numbers from different objects. The 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$; the $L^3 - 1$ counts non-Γ momenta, not cubes. Their agreement is a consistency check on the whole construction, not a restatement.

3.1 The wrapping sheets are cycles, not harmonic

Theorem 2 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x:\, x_m = 0} [x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \operatorname{im} \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s,\ \partial_3 \partial_3^\dagger\, s\rangle}{\|s\|^2} = 2 \qquad \text{exactly, for every } L \geq 2. \tag{7}$$

`check: FINDING: the wrapping sheets are cycles but NOT harmonic`

The distinction matters because H_2 is used in two senses: the quotient in Theorem 2, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in. Nothing downstream breaks — Proposition 26 rests on $B^\dagger w = 0$, which covers all of $\ker \partial_2$ — but a real-space statement phrased for the harmonic subspace does not cover the generators this construction exhibits.

4 Exact all-rank dynamics at second order

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \mathrm{Adj}, \qquad F \otimes F = \Lambda^2 F \oplus \mathrm{Sym}^2 F, \tag{8}$$

with dimensions and quadratic Casimirs

$$\begin{aligned} d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0, \\ d_{\mathrm{Adj}} &= N^2 - 1, & C_{\mathrm{Adj}} &= N, \\ d_{\Lambda^2 F} &= \tfrac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \tfrac{(N+1)(N-2)}{N}, \\ d_{\mathrm{Sym}^2 F} &= \tfrac{N(N+1)}{2}, & C_{\mathrm{Sym}^2 F} &= \tfrac{(N-1)(N+2)}{N}. \end{aligned} \tag{9}$$

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

check: each channel gap is C_F + C_R/2, and the weights sum to one

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 6 (Shared-link channel weights)**.** *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2, for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-$(2,2)$ moment of U.

Second, two such moments settle both families. In the Weingarten calculus, whose asymptotic origin is [3] and whose finite-N closed forms are [9], the order-two values are $\mathrm{Wg}(e) = 1/(N^2-1)$ and $\mathrm{Wg}((12)) = -1/(N(N^2-1))$ — the inverse of the S_2 Gram matrix $\left(\begin{smallmatrix} N^2 & N \\ N & N^2 \end{smallmatrix}\right)$ — the index sums give

$$M_{\text{direct}} = \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \qquad M_{\text{cross}} = \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \tag{10}$$

For the like family, $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU)$ decomposes into $\mathrm{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\text{direct}} \pm M_{\text{cross}})/2$. Normalising,

$$n_{\mathrm{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\mathrm{Sym}^2 F}}{N^2}, \qquad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij}\bar{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1 against $M_{\text{direct}} = N^2$, so $n_{\mathbf{1}} = 1/N^2 = d_{\mathbf{1}}/N^2$ and $n_{\mathrm{Adj}} = 1 - 1/N^2 = d_{\mathrm{Adj}}/N^2$. □

check: the shared-link weights are Weingarten, not an isotropy assumption

Remark 7. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (10) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times N^2-1 is 1 identically in N, so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these are why the all-rank second-order result is self-contained rather than conditional. Appendix B records the index sums.

check: the Weingarten route is independent of the corpus

Remark 8 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [11] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim\rho/(\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N. That is provenance for a chain already checked here, not a second derivation, and it does not check the premise that four channels exhaust the shared-link routes.

check: the published dimension-ratio matrix element is this registry's weight formula

With Theorem 6 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \tag{11}$$

and summing within the two orientation families,

$$A_N = w_{\mathbf{1}} + w_{\mathrm{Adj}} = -\frac{2N^3}{(N^2-1)(2N^2-1)}, \tag{12}$$

$$B_N = w_{\Lambda^2} + w_{\mathrm{Sym}^2} = -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}. \tag{13}$$

check: the four channel weights follow from dimension and Casimir

check: A_N and B_N are the channel sums, not transcriptions

4.2 The shared-link law

Theorem 9 (All-rank shared-link law)**.** *For every integer $N \geq 3$,*

$$H^{(N)}_{\text{eff},-}(k,u) = E_{\text{flat},N}(u) I + t_N u^2 B(k) B(k)^\dagger + O(u^3),$$

with $E_{\text{flat},N}$ momentum independent and

$$t_N = B_N - A_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0.$$

The three bands through second order are $E_{\text{flat},N}(u)$ and $E_{\text{flat},N}(u) + t_N u^2 q(k)$ twice.

Proof. The diagonal and within-plaquette terms are scalars in face-orbital space, and they are scalars for a reason worth naming, because it fails elsewhere in this paper: translations act transitively on the faces of each orientation and the cubic point group permutes the three orientations, so the faces are a *single* orbit and any invariant diagonal is a multiple of the identity. On an open cluster that is false, and Proposition 16 is what replaces it there. Every surviving second-order inter-plaquette process uses one shared link — see Proposition 10 for why the routes that do not are absent from this sector — and its relative orientation is the incidence product appearing in $S = BB^\dagger - 4I$. The contributions (12) and (13) therefore assemble as $d_N I + t_N S$; absorbing $-4t_N$ into the scalar and applying Theorem 1 gives the bands. Positivity: for $N \geq 3$ every denominator factor is positive and $N^2 - 4 > 0$. □

`check: t_N = B_N - A_N`

`check: t_N > 0 for N >= 3`

Two consequences,

$$t_3 = \frac{5}{612}, \qquad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \cdots \tag{14}$$

`check: t_2 = 0 and t_3 = 5/612`

`check: large-N expansion of t_N through 1/N^9`

4.3 The vacuum-mediated route

The proof of Theorem 9 asserts that only shared-link routes survive. That is false in general and true here, for a reason worth stating: it is a charge-conjugation selection rule, and omitting it is a recorded error, not a hypothetical one.

Proposition 10 (Vacuum-mediated route)**.** *The route $\langle p'|V|0\rangle\langle 0|V|p\rangle/E_0$ is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p,-\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such term, while the charge-even shared-link element acquires $1/C_F$:*

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2-5)}{(N^2-1)(2N^2-1)(4N^2-9)}.$$

`check: ell_N = A_N + B_N + 1/C_F, the vacuum-mediated route at every rank`

Remark 11. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

One thing this proposition does *not* do is dispose of the route between non-adjacent pairs. Taken over all pairs it would be a rank-one operator supported at $k = 0$, which is not the twelve-neighbour adjacency ℓ_N multiplies; which contributions survive in H_2 is settled by the linked, vacuum-subtracted prescription of (5), not by anything derived here. What the check above establishes is the all-rank value ℓ_N of the shared-link element, and the disposition of the non-adjacent contributions is prescription rather than result.

4.4 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L) = \begin{cases} 12, & L \text{ even}, \\ 12\cos^2(\pi/2L), & L \text{ odd}, \end{cases} \tag{15}$$

because only at even L does some k_j reach π exactly. The width of the full retained charge-odd manifold is $W^{(-)}_{N,L} = q_{\max}(L)\, t_N u^2 + O(u^3)$, so for even L and in the thermodynamic limit $[u^2]W^{(-)} = 12t_N \sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

`check: the zone maximum of q is 12 only at even L`

The smallest nonzero incidence eigenvalue is $q_{\min} = 4\sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

`check: q_min on the L-torus grid is 4 sin^2(pi/L)`

4.5 The orientation signs are essential

Removing the orientation signs — from the entries of (6) and from the phases, so each $e^{ik} - 1$ becomes $e^{ik} + 1$ — gives an unsigned incidence $\mathcal{N}(k)$ with $A + 4I = \mathcal{N}\mathcal{N}^\dagger$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

This control is not hypothetical: the unsigned spectrum is the charge-even sector's, which Section 6 computes in closed form, and the two spans are the two bandwidths. The signed span is $8 - (-4) = 12$, giving the charge-odd width $12t_- = 5/51$ at $N = 3$; the unsigned span is $12 - (-4) = 16$, giving the charge-even width $16|\ell_3| = 88/153$.

`check: the two band spans ARE the two incidence spectra`

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3, 2, 2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $B = 6$ $SU(3)$ truncation, of dimension 1,590,462. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven — a census result and not the plaquette count. The full E_* eigenspace has dimension 22, split $11_{C=+} \oplus 11_{C=-}$ by tensor-derived charge conjugation, and the delivery removes all 22 from the pseudoinverse; because the eleven columns exhaust the odd half, that agrees with the $Q = 1_- - P_-$ prescription of (5). Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\text{conn}} = K_{LR} - J_L K_L J_L^\dagger - J_R K_R J_R^\dagger + J_F K_F J_F^\dagger, \quad (16)$$

and the same fold applied to the geometry gives G_{conn}. The fold is literal in the sense that no eigenvalue or fitted scalar is subtracted: each source operator is folded in its own coordinates and lifted. The delivery corroborates that internally by applying the same weights history by history over its 1,984 ordered two-step histories, of which 1,192 cancel at weight zero, and the two routes agree to 4.9×10^{-15}.

First order settles itself. On the shell $PVP = I_{11}$, so the first-order term is nonzero but scalar: it shifts all eleven branches equally and creates no branch ambiguity. Its connected fold then vanishes *identically*, and by counting rather than by cancellation — each of the eleven faces is covered exactly once by $\{L, R\}$ once the shared face is restored by F, so every Möbius weight in (16) applied to the identity is $1 - 1 = 0$. Second order is therefore the leading connected term and not a correction to a first-order one.

check: `connected first order vanishes by inclusion-exclusion, not by cancellation of numbers`

At second order the result is

$$K^{(2)}_{\text{conn}} = \frac{5}{612} G_{\text{conn}} + D, \quad (17)$$

with D diagonal and delivered. The geometry alone fixes the *splitting* inside each block and nothing else: G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), so each pair splits by $2 \times 5/612 = 5/306$ about its own diagonal entry. Given D, which Appendix F prints, the spectrum is then closed form, $\{0, (-\frac{15}{4})^2, (-\frac{34}{9})^4, (-\frac{129}{34})^4\}$ — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind — supplied to no builder, graph fit, branching rule or validator, with the payload hashes frozen before any comparison — and this repository re-thresholds those gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects: on a 66-dimensional held-out word space the correct second-order residual falls with log-log slope 2.90 while the sign-reversed control falls with slope 2.00, a minimum error ratio of 325. What all this establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration.

check: `the connected two-cube geometry has exactly four cross-cell pairs, each -1`

check: `the B=6 connected kernel is (5/612) G_conn + diag, with the certified spectrum`

check: `the certificate's own gates pass, target-blind, with the wrong-sign control rejected`

Resolving (17) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

ρ	$\mathbf{1}$	$\mathbf{3}$	$\bar{\mathbf{3}}$	$\mathbf{6}$	$\bar{\mathbf{6}}$	$\mathbf{8}$
c_ρ	$\frac{1}{12}$	$-\frac{1}{12}$	$-\frac{1}{12}$	$-\frac{1}{9}$	$-\frac{1}{9}$	$\frac{16}{51}$

(18)

check: `the B=6 six-channel census sums to the registry's own t_3 = 5/612`

That these six sum to $5/612$ is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (18) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.2×10^{-15} per channel — 2.1×10^{-14} across the delivery's reconstructed matrices as a whole — and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 6.

Reported computational result 12 (The census is the Weingarten ledger)**.** At $N = 3$ the six reconstructed link-irrep coefficients of (18) are the four fusion-channel weights (11) individually:

$$c_{\mathbf{1}} = -w_{\mathbf{1}}, \qquad c_{\mathbf{8}} = -w_{\text{Adj}},$$

$$c_{\mathbf{3}} = c_{\bar{\mathbf{3}}} = \tfrac{1}{2} w_{\Lambda^2 F}, \qquad c_{\mathbf{6}} = c_{\bar{\mathbf{6}}} = \tfrac{1}{2} w_{\text{Sym}^2 F}.$$

Substituting (9) into (11) at $N = 3$ gives $w_{\mathbf{1}} = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\text{Sym}^2 F} = -2/9$; comparison with (18) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

check: `the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel`

One further feature of the census is worth separating out, because it is the paper's central architecture measured rather than assumed. The six link-resolved matrices are not merely six numbers that sum correctly: each is *separately* proportional to the same incidence Gram. The eleven oriented plaquettes of the prism have exactly 56 ordered adjacent pairs, each sharing one link with signed overlap ± 1; the delivery reports a constant channel-to-geometry ratio on all 56 of them, for every one of the six channels, with worst residual 2.2×10^{-15}. Colour dynamics fixes a number per channel and the cellular boundary operator fixes the matrix; if the two mixed, the ratio would be constant for the total and not for the parts.

check: `every shared-link channel is separately proportional to the geometry, on all 56 pairs`

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2. Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $B = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate — the two orientations of the shared link weighted equally — which is measured here on the 1,590,462-state space rather than assumed.

Conjecture 13 (All-rank even split)**.** For every $N \geq 3$ each like-family fusion weight splits evenly between an irrep and its conjugate in the link-resolved census.

This is a conjecture and nothing here promotes it. Falsifying it means a link-resolved census at some $N \geq 4$ in a truncation that still retains every adjacent shared-link channel; at $N = 4$ the endpoint budgets make that $B \geq 33/4$, and no such construction has been run.

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3. An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation: the sextet needs exactly 6 and the adjoint 17/3, so both survive $B = 6$ and neither survives $B = 4$.

check: `B=6 retains every adjacent shared-link channel; B=4 provably cannot`

Corollary 14 (What a link cutoff drops)**.** *Retaining only the singlet and $\Lambda^2 F$ routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_{\mathbf{1}} = -\tfrac{1}{12},$$

opposite in sign to t_3 and 10.2 *times larger in magnitude. What is dropped is*

$$w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = \tfrac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$. This is arithmetic on (11) *alone.*

The published $p + q \leq 1$ link cutoff retains exactly those two routes, so $-1/12$ is its projected hopping and $14/153$ is the counterterm that completes it — not $5/612$, which would double-count what it already carries. On the two-cube space the first of those two numbers is delivered rather than inferred: a separately sealed $B = 4$ run on the identical geometry gives $K^{(2),B4}_{\mathrm{conn}} = -\frac{1}{12} G_{\mathrm{conn}} + D_{B4}$ outright, and the Casimir budgets of the previous paragraph say without reference to any census that $B = 4$ can reach only $\{\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}\}$. Conditional on (18), the same split of the census reproduces it, $c_{\mathbf{1}} + c_{\mathbf{3}} + c_{\bar{\mathbf{3}}} = -1/12$, and gives the omitted half as $c_{\mathbf{6}} + c_{\bar{\mathbf{6}}} + c_{\mathbf{8}} = 14/153$.

check: `the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel`

check: `FINDING: the T1 link cutoff reverses the sign of t_3, and 14/153 is what it omits`

Remark 15. Two single-cube reconstructions bracket the same statement. A full $B = 4$ open cube assembled from the published `ymcirc` $d = 3$, $B = 4$ plaquette table [12] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11}; a $B = 6$ cube reproduces $+5/612$ and the inverse ordering to 8.4×10^{-15}. Both locate the missing weight at the same 14/153.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage — and the two-cube build draws its Clebsch–Gordan amplitudes from the same hash-pinned `pyclebsch` archive as the $B = 6$ cube, so that agreement is a cross-check of assembly and not of the amplitudes. The published closed form [11] is d_ρ / N^2 — the very numerator of (11). This is a cross-check of assembly, not a third confirmation of t_3; the exact rational identity is Corollary 14, which needs no tolerance. The $p + q \leq 1$ nomenclature and the finite-rank matrix element are both from [11], and single-cube $SU(N)$ diagonalisation in this style goes back to [8].

A counterterm added to a published T_1 Hamiltonian is therefore 14/153 and not 5/612, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight $w_{\mathbf{1}}$, and the cutoff value is a *difference* of two weights, not a weight.

check: `FINDING: a full T1 = B = 4 cube Hamiltonian reproduces -1/12 and the reversed shell`

check: `FINDING: the B = 6 cube flips the sign back to +5/612, closing the decisive test`

The $B = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is 39/68 against the bridge's 11/34, differing by exactly 1/4, the local same-face sextet route, which first enters at cutoff 20/3 — integer label $B = 7$, so any cutoff above 20/3 already carries it.

`check: the B = 6 scalar misses the bridge's by exactly the same-face sextet route`

5.2 What the cutoff moves, and where

The connected diagonal D is eleven numbers in both deliveries and is treated as a remainder in both. It is not eleven numbers. The open prism has a symmetry group — the cell exchange $x \mapsto 2-x$, the two transverse reflections and the $y \leftrightarrow z$ swap, of order 16 — and D is constant on its orbits, which are $8+2+1$: the eight side faces, the two end caps, the shared face.

Proposition 16 (Where the truncation lives)**.** *The eight-element orbit is exactly the support of the four cross-cell pairs, and it is the only orbit whose diagonal value differs between the two truncations: the end caps carry* $-15/4$ *and the shared face* 0 *at both* $B = 4$ *and* $B = 6$*, while the side value moves from* $-7/4$ *to* $-2317/612$*. Together with the four cross-cell blocks carrying the whole off-diagonal change, the entire* $B = 4 \to B = 6$ *difference is confined to the faces that carry connected transport.*

`check: the connected diagonal is orbit-constant, and only the transporting orbit moves`

This also says what the nonuniform D is. Where the faces of a complex form a *single* orbit — the closed cube of the next subsection, the periodic torus of Theorem 9 — orbit-constancy forces the diagonal to be scalar, and the operator is exactly $\alpha I + tG$. The prism has three orbits because it has a boundary. D is the open boundary showing up in the diagonal, not a failure of the incidence form.

5.3 The one-cube shell, and why its flat level cannot move

Those two single-cube reconstructions are usually read as numerical cross-checks. They are more than that, because the object they diagonalise is the same chain complex. The six faces of a cube are a closed surface, so in the outward orientation $\ker \partial_2$ is one-dimensional and spanned by the sum of all six — the fundamental class, $b_2(S^2) = 1$. That is Theorem 2 on a complex with no three-cells, where the $L^3 - 1$ boundaries are absent and only the class survives.

Proposition 17 (One-cube shell)**.** *On the closed cube surface* $G = \partial_2^{\mathsf{T}} \partial_2$ *commutes with all 24 proper rotations. The combinations* $b_{+i} - b_{-i}$ *carry the defining vector representation and the combinations* $b_{+i} + b_{-i}$ *the permutation representation of the three unordered axes, so with the charge-odd inversion* $Pb_{+i} = -b_{-i}$ *the shell is*

$$A_1^{--} \oplus T_1^{+-} \oplus E^{--},$$

with G*-eigenvalues* 0, 4, 6 *and multiplicities* 1, 3, 2*. Consequently* $\alpha I + tG$ *has levels* α, $\alpha + 4t$, $\alpha + 6t$*.*

Proof. The rotations permute the six outward faces, and G is built from the boundary map, so it commutes with the permutation action. On the antisymmetric combinations that action is the defining representation by inspection — $b_{+i} - b_{-i}$ transforms as e_i — and on the symmetric combinations it is the permutation representation of $\{\pm e_i\}$, which is $A_1 \oplus E$. Under $Pb_{+i} = -b_{-i}$ the antisymmetric sector is parity even and the symmetric one parity odd. In these sectors G is $4I$ and $4I - 2(J - I)$ respectively, of spectra $\{4, 4, 4\}$ and $\{0, 6, 6\}$; since the three isotypic components have distinct dimensions the eigenvalues attach to them uniquely. $\square$

`check: the one-cube shell is A1 + T1 + E, and its flat level is the S^2 fundamental class`

Three things follow. The cubic labels are derived here rather than assigned: both deliveries assert them and neither builds the symmetry matrices. The A_1^{--} level is Proposition 26's mechanism on a second complex — $Gw = 0$ there, so that level sits at α for every t, and no truncation, no channel and no coefficient in dispute anywhere in this paper can move it. And the reversal the two runs report is therefore a statement about the sign of one number rather than about the shell as a whole: at $B = 6$ the levels are $\{39/68, 371/612, 127/204\}$ and at $B = 4$ $\{37/12, 11/4, 31/12\}$, in the opposite order, with A_1^{--} in each case sitting exactly at the scalar. The relative shell $\{0, 4t, 6t\}$ is $\{0, 5/153, 5/102\}$ at $B = 6$.

The finite-volume fingerprint of the 28 August finite-N bridge certificate — a different release, and named here because a reader tracing it to the two-cube one would find nothing — is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence spectrum is $\{0^{29}, 3^{12}, 6^{24}, 9^{16}\}$, and $29 = L^3 + 2$ is Theorem 2.

`check: the certificate's finite-volume fingerprints, and 29 = L^3 + 2 is the Lean cycle count`

Remark 18 (Scope)*.* Everything in this section is second order in u, finite volume, strong coupling, one sector. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The 1,590,462-state construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B-truncated and not proved cutoff-stable. The held-out finite-u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space with the

Q_1–Q_1 magnetic block set to zero, not the full Hamiltonian, and no radius-three stability claim was executed. The two branches are not equally covered: at $B = 4$ the delivery reports a direct diagonalisation of the complete 4,180-dimensional charge-odd block at four small couplings, two of them held out, with the residual after $E_* + u + u^2 \operatorname{eig} K_2$ scaling as u^3; at $B = 6$ no such diagonalisation exists and the 66-dimensional star stands in for it. That $B = 4$ report carries no check here — its certificate did not travel — and is recorded as the delivery's rather than as this paper's. No external group has reproduced the release.

6 The charge-even sector in closed form

The unsigned incidence $\mathcal{N}(k)$ of Section 4 is not a hypothetical control: it is the charge-even sector's Bloch symbol. Write

$$\hat{a}_m = 2 + 2\cos k_m = |1 + e^{ik_m}|^2, \qquad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4\sin^2(k_m/2)$ used in the charge-odd shape basis (20) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 19 (Charge-even Bloch cubic). *$p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{N}\mathcal{N}^\dagger$ is*

$$\mu^3 - 2p\,\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 \;=\; \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the charge-even adjacency eigenvalues $\lambda = \mu - 4$. The charge-odd counterpart of the same computation is $\mu(\mu - q)^2$.

```
check:    the C-even characteristic polynomial is mu(mu -
p)^2 = 4 a_1 a_2 a_3
```

```
check: p + q = 12:  one zone function runs both sectors
```

The closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

```
check:       the Bloch cubic IS the finite L = 3 and L = 4
plaquette spectrum, exactly
```

Proposition 20 (Range and attainment). *$\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.*

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \le 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $f(16) = 16(16 - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3 \ge 0$ because $p \le 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 \le 64$. Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ. Equality at the bottom is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. ☐

```
check:   the C-even range [-4, 12] is exact; the top only
at Gamma, the floor on three planes
```

```
check:   the C-even band touches its floor exactly on the
three planes k_j = pi
```

Remark 21. A flat charge-even branch would need $f(\mu_0) = 0$ identically in k, which the constant term already forbids. The charge-even kernel is three measure-zero planes against the charge-odd kernel's whole band: the two sectors are two readings of one geometric object, and only the signed one is flat.

```
check:   the C-even band touches its floor exactly on the
three planes k_j = pi
```

```
check:        the C-even spectra at the four high-symmetry
momenta
```

6.1 One assembly, both sectors, both orders

The $r = 2$ content below is all-rank and checked. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

The scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\mathrm{tower}_{r,s}$ for the within-plaquette term in canonical u,

$$E_s(\lambda, r) = \mathrm{tower}_{r,s} + 12\,\mathrm{leak}_{r,s} + \lambda\, t_{r,s}, \tag{19}$$

with $\lambda \in \{-4, 8\}$ in the charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the charge-even sector, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta(3u/2)$, so (19) takes no unregistered input. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link. The two towers are different objects and both are needed: the charge-odd one contributes $8/3 + u + u^2/2 + 7u^3/32$ and the charge-even one $8/3 - u + (13/20)u^2 + (101/200)u^3$, each of them $4\Delta(3u/2)$ on its own sector's series, so $\mathrm{tower}_{2,-} = 1/2$ against $\mathrm{tower}_{2,+} = 13/20$ and $\mathrm{tower}_{3,-} = 7/32$ against $\mathrm{tower}_{3,+} = 101/200$.

check: one assembly formula gives every registered band value

check: one assembly formula gives every C-even value at both orders

check: the plaquette graph is 12-regular and two faces share at most one link

Because $\mathrm{leak}_{r,+} = t_{r,+}$ at both computed orders, the whole charge-even sector collapses to $\mathrm{tower}_{r,+} + (12+\lambda)\, t_{r,+}$, a one-parameter family in the charge-even tower. That equality is verified and unexplained: at third order the charge-odd leakage separates from the charge-odd hopping while the charge-even identity survives. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

check: both declared coincidences, checked: one is ell_N at all ranks, the other is bare

6.2 Where the rest frame is not blind

Proposition 22 (Charge-even Γ splitting)**.** *At Γ the charge-even spectrum is $\{12, 0, 0\}$, so the A_1^{++} level and the E^{++} doublet are distinct, and*

$$E(A_1^{++}) - E(E^{++}) = 12\, t_{r,+}$$

exactly: $-22/51$ at $r = 2$ and $-6335/20808$ at $r = 3$.

check: the C-even Gamma point pins t_+, exactly where no C-odd Gamma datum can

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned spectrum has two levels, so the charge-even rest frame can. The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — $22/459$ at second order and $6335/187272$ at third.

check: the C-even curvature is (4/3)|t_+| at both orders, and isotropic

At both computed orders the charge-even bandwidth is $16|t_{r,+}|$ and the charge-odd manifold width $12|t_{r,-}|$. At any order at which the correction stays in the incidence algebra — which Proposition 26 warns is not automatic — the second-order ratio is $4(3N^2-5)/[3(N^2-4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from $88/15$ at $N = 3$ towards 4. So at second order the charge-even band is the wider one at every rank, and that asymmetry is a channel statement, not a geometric one.

check: the C-even bandwidth is 16|t_+| at every order; the C-odd manifold width is 12|t_-|

check: at N = 2 the C-odd hopping vanishes and the C-even one does not

7 $SU(3)$ through third order

The *third-order* content of this section is conditional on supplied exact-arithmetic ledgers. Its second-order content is not: leak_2 is the all-rank ℓ_N of Proposition 10 at $N = 3$, and the tower is the certified coupling conversion. The within-plaquette charge-odd tower contributes $8/3 + u + u^2/2 + 7u^3/32$.

Reported computational result 23 (Third-order ledger)**.** The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \qquad \mathrm{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12\,\mathrm{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

check: d_3 = 7/32 + 12*leak_3 - 4*b_3

check: leak_3 is assembled from the domino diagonal and the vacuum piece

Conditional corollary 24 (Third-order factorisation)**.** If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\mathrm{eff},-}(k,u) = E_{\mathrm{flat}}(u) I + \tau(u) B(k) B(k)^\dagger + O(u^4),$$

$$\begin{aligned} E_{\mathrm{flat}}(u) &= \tfrac{8}{3} + u + \tfrac{11}{306}u^2 - \tfrac{109151}{249696}u^3, \\ \tau(u) &= \tfrac{5}{612}u^2 + \tfrac{1975}{124848}u^3. \end{aligned}$$

check: E_flat and t(u) carry the ledger coefficients

The second-order diagonal is the $\lambda = 0$ value $\frac{1}{2} + 12\,\mathrm{leak}_2 = 7/102$ with $\mathrm{leak}_2 = -11/306$; subtracting $4t_-$ gives the flat scalar $11/306$ at $\lambda = -4$.

check: d_- = 1/2 + 12*leak_2, and leak_2 = -11/306

Remark 25 (What the external comparison can and cannot do). Hamer's 1989 rest-frame series [7] agrees with E_{flat} at every shared order. It cannot test τ. The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ-point datum is blind to it — the charge-even rest frame is not, by Proposition 22; the one-parameter family $(b_3 + \varepsilon,\ \mathrm{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\,\mathrm{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (19). Proposition 22 is the charge-*even* analogue and separates leak_+ from t_+; it says nothing about b_3 or leak_3.

check: FINDING: no Gamma-point datum can constrain the hopping

8 What the homology protects

Proposition 26 (Boundary-factorised rigidity)**.** *If an order-r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 1. The middle term is annihilated whatever M_r is, which is the content: the protection belongs to the boundary operator and not to the dynamics. □

check: `a boundary-factorised correction shifts the carrier without dispersing it, for generic M`

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\mathrm{End}(C_2)$, and a general higher-order correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \tag{20}$$

with $e_2 = \sum_{i<j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4\sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 1, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger H_4 w)/q$ and the numerator, not ε_4, is the polynomial.

check: `the carrier projection is where the 1/q comes from`

check: `clearing the denominator reproduces the five-element numerator basis`

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient C_{shp}. This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What it does is state precisely where the disagreement lives, because that geography is now checked and it is what a decisive calculation must target.

The two records do agree on the axial datum, but the corpus's own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13}. The sealed core holds internally — α_{new} and $4A_{\mathrm{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

check: `v10a.26 A, B, D match the sealed rationals within 2.3e-13`

check: `FINDING: alpha_new falls outside the corpus's own 2.3e-13 bound`

9.1 The obstruction is polynomial

Theorem 27 (Obstruction certificate)**.** *The two recorded off-axis coefficients enter (20) only through the shape $4e_2/q$, so the two records' carrier numerators differ by $4\Delta_C e_2$ and their bands by $\Delta_C\,(4e_2/q)$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the* zero polynomial *there and both differences vanish identically. Consequently no Γ-point or axial datum distinguishes the two records, at any precision. Off axis they do differ: $e_2(M) = 16$ and $e_2(R) = 48$, so the numerators differ by $64\Delta_C$ and $192\Delta_C$ and the bands by $8\Delta_C$ and $16\Delta_C$.*

check: `FINDING: the retained Gamma/axis data cannot identify C_shp`

check: `Phi_C vanishes at Gamma along every direction`

Remark 28. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

check: `the crosswalk is exactly scalar on the momentum axes`

On the axial cut itself the invariants collapse to $q = L(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$, and dividing the raw cube-boundary numerator $(\alpha_N/4)L^2$ by $\|w\|^2 = q = L$ leaves a single power of L with coefficient $\alpha_3/4 = 5/48$.

check: `on an axial cut the mixed invariants vanish and the norm divides`

9.2 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The witness below is not itself a candidate for the physical coefficient — it is the balanced ϵ-free continuation of the all-rank shape family, and ϵ-free is precisely what the physical value is not. Its job is to make the $4\Delta_C e_2$ family concrete with an exactly known shift.

Proposition 29 (Explicit second witness)**.** *There is a value C_{alt} with $C_{\mathrm{alt}} - C_{\mathrm{old}} = 25/1024$ exactly, appearing verbatim in the pinned source edition, whose induced band separations are 0 at X, $25/128$ at M and $25/64$ at R. It is identical to C_{old} on every retained datum and distinct from it off-axis.*

check: FINDING: an explicit second witness C_alt exhibits the C2 non-identifiability

check: the C_alt witness IS the balanced eps-free continuation, and Delta_C/(A/2) = 15/32

9.3 Where the coefficient lives

The four checkpoint deltas are the Γ-subtracted carrier corrections $\Delta_s = \varepsilon_4(k_s) - \varepsilon_4(0)$ at $k_X = (\pi, 0, 0)$, $k_M = (\pi, \pi, 0)$, $k_P = (\pi, \pi/2, 0)$ and $k_R = (\pi, \pi, \pi)$ — P is the off-axis, off-corner point that makes the system invertible, with $a = (4, 2, 0)$. In the shape basis they are

$$\begin{aligned} \Delta_X &= 4A, \\ \Delta_M &= 8A + 16B + 8C, \\ \Delta_P &= 6A + 8B + \tfrac{16}{3}C, \\ \Delta_R &= 12A + 48B + 16C + \tfrac{16}{3}D, \end{aligned} \tag{21}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

check: checkpoint values at X, M, P, R

check: the four extraction formulas invert the ansatz

check: X is blind to B, C, D | it fixes A alone

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15}, as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q^{(4)}_{\text{band}}$ and $m^{(4)}_{\Gamma}$ are differently anchored coordinates and not rival estimates.
2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.
3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h, and it sets both shapes rigidly: $A_{\text{normal}} = -h$ while the raw normal amplitude is $2h$, which in the shape normalisation of (20) is $C_{\text{normal}} = h/2$, so

   $$C_{\text{normal}} = -\tfrac{1}{2} A_{\text{normal}} \quad \text{exactly.}$$

   (The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A-carrying sector; no redistribution reaches the gap.
4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

check: the shape fit's C row sums to zero, so no Gamma-anchor error can move C_shp at all

check: a translation-local scalar shift changes nothing observable

check: FINDING: C_shp is carried by 6 of 189 records, not spread across the kernel

check: FINDING: A = 5/48 pins the normal sector's whole C4 contribution

check: C_normal = -A_normal/2: the agreed axial coefficient pins the normal channel

check: the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A-carrying sector rather than by redistribution, which is a constraint on *any* candidate, including both recorded ones. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.4 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ-point scalar and emits no kernel block at all, and by Theorem 27 Γ-point data are structurally incapable of constraining C_{shp}. This is a statement about the instrument, made by reading it, not a report of a failed run.

check: FINDING: the marked-cluster engine emits the Gamma scalar only | a completed 609-sweep cannot decide C_shp

9.5 The tier collapse, as two integers

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a-degree three, because the projection contributes one further power of a: for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\mathrm{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3. An earlier degree-bound mechanism proposed for the collapse was therefore retracted. What is checked is finer: every degree-three block amplitude is an integer multiple of one raw record weight x, and

$$B = 0:\ (+1, +1, -2)\cdot x, \qquad D = 0:\ (-3, +6, -3)\cdot x$$

over the displacement shells. The kernel does populate the forbidden tier; the dynamics cancels it with small integers. Any mechanism must reproduce those two vectors.

check: `RETRACTED: the vertex count does NOT forbid B_shp and D_shp`

check: `FINDING: the tier collapse is two integer cancellations at record level`

check: `the vanishing coefficients are exactly the degree-3 ones`

9.6 A near-Γ statement that survives the dispute

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. With Jordan's inequality $q(k) \geq (4/\pi^2)|k|^2$ on the whole zone — tight at $k = 0$ and at the zone corner — and $\tau(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are dominated by the second-order dispersion outside a ball of radius Ku with $K = (\pi/2)\sqrt{W_4/(\theta t_3)}$. The two records give $K = 17.04$ and $K = 23.66$ — floating point, since one of the two bandwidths exists only as a float — and the larger bounds both, so the exclusion radius can be stated now. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and the excluded zone fraction grows as u^3.

check: `Jordan bound q(k) >= (4/pi^2)|k|^2 holds on the whole zone`

check: `the criterion survives C2: K depends on the kernel only through sqrt(W_4)`

check: `the statement is non-vacuous only below an explicit coupling`

10 External comparisons

External results count here because they are *independent* — produced without knowledge of this program — not because they are published. Each is an assertion until something checks it.

Hamer 1989, both sectors. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2$, $4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12}. The 0^{++} series matches the charge-even A_1^{++} Γ-point coefficients at x^2 and x^3, with the sign structure ($a_1 = -1$ in the scalar channel where the carrier's is $+1$) reproduced; the E^{++} doublet is not constrained by it, which is why Proposition 22 needs the splitting and not the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer's normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [2]; that paper has not been read here, so the disagreement is recorded and not adjudicated.

check: `Hamer's 1+- series matches the C-odd Gamma-point coefficients through x^3`

check: `Hamer's 0++ series matches the C-even Gamma-point coefficients through x^3`

check: `the m_n = 2^(n-1) a_n bridge is the x = 2u conversion`

check: `the Hamer table is pinned, and the a_4 agreement is primary-source`

The string tension. The Kogut–Pearson–Shigemitsu $SU(3)$ table [4] — read here from the 1980 preprint, the journal pages themselves being unread — equals this program's certified σ series exactly, $2^{n-1}t_n = \sigma_n$ for $n = 2, \ldots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3, including the discriminating u^3 coefficient.

check: `the KPS 1980 string-tension table equals the certified sigma series EXACTLY`

check: `sigma_n^phys = (-1)^n sigma_n^raw, and C5 is the n = 3 case`

check: `the ratio and sigma series reproduce E_flat exactly`

Euclidean series. Two printings of the same Münster/Seo series — the NPB 205 (1982) erratum row and Smit's Table 1 scalar column, which credits them — agree rational for rational through u^8. That is agreement transcription for transcription, not two independent computations, and it is recorded as such. One shift is recorded and not adjudicated: Münster's 1985 table [5] differs from the 1982 erratum row at eighth order by exactly -96, a fourth correction already in print by 1983 that no erratum records. This is a Euclidean record; no Hamiltonian comparison is drawn from it, and none is permitted.

check: `the errata-resolved Euclidean series is doubly sourced, transcription for transcription`

check: `FINDING: Munster's 1985 table shifts his 1982 erratum at eighth order`

The overlap obstruction. Schierholz [6] measured in 1988 that at $\beta = 5.9$ a bare local operator carries only a couple of percent of the physical glueball state, and gave the scaling a^5 for a local dimension-four operator — which is why that percentage needs its coupling quoted alongside it. That makes the smeared basis structural rather than a convenience: better statistics cannot recover a power law. It is the published, independent form of the obstruction this program records as its own largest debt.

check: `the overlap obstruction was published in 1988, and it scales`

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [13], whose Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [10]. The rule is enforced by the literature validator rather than recorded in prose.

check: `a cross-regime paper never supplies a value`

11 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-}; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L, but its nearest incidence separation is $\Delta_L(u) = 4\tau(u)\sin^2(\pi/L) + O(u^4)$, which falls as L^{-2} and so provides no volume-uniform spectral isolation.

check: `Delta_L = 4 tau(u) sin^2(pi/L) is positive and falls as L^-2`

Three crossings are named explicitly, and none of them is made here.

- **Finite lattice to infinite volume** requires an inhomogeneous Wilson free-energy bound with a useful $\log K_\alpha$ and a source-radius reduction. Both are open, and are carried in the accompanying gap register as `G17`.
- **Protected level to particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the carrier with the transfer-matrix spectrum, and multi-plaquette survival of the protected level (`G18`). The published overlap measurement of Section 10 is why this theorem must live in the smeared basis.
- **Lattice to continuum** (`G19`) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

The second and third are the only route from the statements above to any sentence containing the words "mass gap", and nothing in this edition shortens that route.

Two premises earlier in the paper are arguments in prose rather than results, and both are load-bearing. That the range of P_- is exactly the charge-odd one-plaquette span for $L \geq 3$ is argued in Section 3 from the link count of a reduced fundamental loop. That the four fusion channels exhaust the second-order shared-link routes is assumed in Theorem 9; Theorem 6 fixes the weight of each channel and does not establish that the list is complete. Neither has a check.

12 Reproducibility

A portable standard-library Python program, `paper/verify_core.py` in the accompanying repository (the root-level file of the same name belongs to the earlier flat-band manuscript and is a different, smaller program), runs twenty-three checks in nine claim groups: the rational coefficient ledger; the Weingarten derivation of Theorem 6 by explicit index summation; the incidence identity, *decided* in exact Gaussian-rational arithmetic at rational torus points rather than checked to a residual; the chain condition $\partial_2\partial_3 = 0$ over $\mathbb{Z}$ and the finite-torus ranks with an explicit kernel basis, together with the closed cube surface of Proposition 17; the Rayleigh quotient of the wrapping sheets; the two-cube connected geometry rebuilt from oriented cell boundaries, its exact $B = 6$ and $B = 4$ spectra, and Reported result 12 channel by channel; the charge-even cubic of Theorem 19 with $p + q = 12$ and the four high-symmetry spectra; the checkpoint algebra of (21) together with the vanishing of e_2 on every axial cut; and the coupling convention, including the 16 and 64 by which a 4^r rescaling misses.

Every value in it is a Fraction: no float is constructed anywhere, so "agreement" there always means equality. It has no dependencies and runs in about half a second.

Nothing in verify_core.py assumes or decides a value of C_{shp}. Its fourth-order group is the obstruction as arithmetic: the extraction formulas, the blindness of X, and the identical vanishing of e_2 on axial cuts.

Every theorem, proposition, corollary and reported result above names the machine check that establishes it, and the appendix tables name checks too. Three displayed objects deliberately carry none, because none is a result of this program: the dimension and Casimir table (9), which is standard $SU(N)$ representation theory; the boundary formulas of Appendix E, which are definitions — what is checked there is that they compose to zero; and Table 1, which is provenance about a commit rather than a claim about the physics, and whose rows are re-measured by the commands printed beside it. The checks live in the accompanying repository, where workhouse verify --only '<name>' reruns any one of them alone, with its numbers, in about a second. That device is itself checked: a registered invariant reads every pinned edition in paper/ — this one included — resolves every printed label against the live registry, and fails if one is renamed, deleted, or left red. The guard is the reason this edition's labels can be trusted at all: the same invariant read only the previous edition until this one landed, and a label that had drifted passed unnoticed until it was made to read both.

This edition pins commit 5ca91a92ac3b15b4a9ef5138c408f08947b16ffa of the accompanying repository — the commit at which every check name printed above resolves, and the last one before this file itself lands. It carries the counters in Table 1. They describe *that* commit and no other, and each row has its own command: after git checkout 5ca91a9, make verify gives the T_1 and T_2 rows and make check the test count, while the T_0 row is the declaration scrape that workhouse certified prints — make lean compiles the tree and reports no count. They are provenance rather than independent evidence; the argument is above, and the checks are the checks.

A smaller layer is proof-checked rather than merely re-derived. The Lean development compiles the rank law with denominators cleared, $t_3 = 5/612$ and $t_2 = 0$, the third-order ledger identity, the axial law at every exceptional rank, the cycle count $(L^3 - 1) + 3 = L^3 + 2$, the four checkpoint-extraction formulas and the deltas they invert, and four statements about the charge-even cubic including the identity that its middle coefficient is p^2. It is exact-rational and polynomial algebra only: no perturbative derivation, no operator theory, and nothing that bears on C_{shp} appears there, because none of that reduces to rational arithmetic.

Machine checks certify algebra relative to stated definitions and inputs; they do not replace a derivation, and they do not certify an uncompleted off-axis contraction — which is why Section 9 stops where it does.

Layer	Count
T_0 — Lean 4 theorems, no sorry	40
T_1 — exact re-derivations	183
T_2 — numerical, within a stated tolerance	47
Repository tests	475

Table 1: Counters at the pinned commit. The T_0 row counts theorem declarations with no sorry, which is a scrape of the tree; make lean is what compiles them, and the standard the tree is held to is that every axiom footprint lies in {propext, Classical.choice, Quot.sound}. That compilation was last run elsewhere: the environment this edition was typeset in carries no Lean toolchain, and the row is not asserted as re-measured here.

A Channel weights in closed form

Substituting (9) into (11),

$$w_{\mathbf{1}} = -\frac{2}{N(N^2-1)}, \qquad w_{\Lambda^2 F} = -\frac{N-1}{(N+1)(2N-3)},$$
$$w_{\text{Adj}} = -\frac{2(N^2-1)}{N(2N^2-1)}, \quad w_{\text{Sym}^2 F} = -\frac{N+1}{(N-1)(2N+3)}.$$

Their family sums are (12) and (13), and the difference has numerator $2N(N^2-4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once.

check: the four channel weights follow from dimension and Casimir

At $N = 3$: $w_{\mathbf{1}} = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\text{Sym}^2 F} = -2/9$, which are the four values Reported result 12 matches against (18).

B The Weingarten index sums

With $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2-1))$, and σ, τ running over S_2,

$$\langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle = \sum_{\sigma,\tau} \text{Wg}(\sigma\tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}.$$

Summing the delta products over all N^4 index quadruples,

$$M_{\text{direct}} = \text{Wg}(e)(N^4 + N^2) + 2\,\text{Wg}((12))N^3 = N^2,$$

$$M_{\text{cross}} = 2\,\text{Wg}(e)N^3 + \text{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$.

`check: the shared-link weights are Weingarten, not an isotropy assumption`

The related fourth moment $\int |U_{11}|^4\, dU = 2/[N(N+1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

`check: SU(3) Weingarten values follow from the general formula`

`check: the fourth moment integral |U_11|^4 = 1/6 at N = 3`

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

`check: the manuscript's SU(3) ledger is this registry, value by value`

The u^1 row is $SU(3)$-only rather than an $SU(3)$ specialisation: it is $-\langle p,-|V|p,-\rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. Of the rest, the u^0 row and the u^2 $BB^\dagger$ coefficient specialise all-rank statements ($2C_F$ and t_N); the u^2 flat scalar inherits the $SU(3)$ tower term through (19); and the three u^3 rows are $SU(3)$-only and conditional on the supplied domino ledger, exactly as Section 7 says.

D The charge-even ledger

In the same convention, with $\lambda \in \{12, -4\}$ and (19):

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16\lvert t_+\rvert$	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	on-site diagonal ($\lambda = 0$)	$52163/260100$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

`check: one assembly formula gives every C-even value at both orders`

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it `bandmin` at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

`check: FINDING: the certificate key 'bandmin' holds the band MAXIMUM, at both orders`

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned}\partial_3[x; 1, 2, 3] = [x + e_1; 2, 3] &- [x; 2, 3] \\ &- [x + e_2; 1, 3] + [x; 1, 3] \\ &+ [x + e_3; 1, 2] - [x; 1, 2].\end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2\partial_3 = 0$ over $\mathbb{Z}$.

`check: d_2 d_3 = 0 on the built complex`

On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\operatorname{rank} \partial_3 = L^3 - 1$.

`check: rank d_3 = L^3 - 1 on the built complex`

F The two-cube geometry

The eleven oriented plaquettes of the $(3, 2, 2)$ prism, by base vertex and ordered plane axes, are

$$\begin{aligned}&(000; 12),\ (000; 13),\ (000; 23),\ (001; 12),\\ &(010; 13),\ (100; 12),\ (100; 13),\ (100; 23),\\ &(101; 12),\ (110; 13),\ (200; 23),\end{aligned}$$

with left, right and shared-face index sets $I_L = (0, 1, 2, 3, 4, 7)$, $I_R = (5, 6, 7, 8, 9, 10)$, $I_F = (7)$. Building $G = BB^\intercal$ from the oriented boundaries and applying the geometric Möbius fold (16) leaves exactly four nonzero off-diagonal pairs, $(0, 5), (1, 6), (3, 8), (4, 9)$, each equal to -1: four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form once the diagonals below are supplied. The two connected diagonals are

$$\begin{aligned}D_{B6} = \tfrac{1}{612}\operatorname{diag}(&-2317, -2317, -2295, -2317, -2317,\\ &-2317, -2317, 0, -2317, -2317, -2295),\\ D_{B4} = \operatorname{diag}(&-\tfrac{7}{4}, -\tfrac{7}{4}, -\tfrac{15}{4}, -\tfrac{7}{4}, -\tfrac{7}{4},\\ &-\tfrac{7}{4}, -\tfrac{7}{4}, 0, -\tfrac{7}{4}, -\tfrac{7}{4}, -\tfrac{15}{4}),\end{aligned}$$

so both spectra are one line of arithmetic: a pair block contributes $D_{ii} \mp c$ twice and each isolated direction its own D_{ii}. The $B = 4$ comparator on the same geometry gives $\{0, (-\frac{15}{4})^2, (-\frac{11}{6})^4, (-\frac{5}{3})^4\}$: the paired levels sit *above* the isolated $-15/4$ pair, where at $B = 6$ they sit below it. The three isolated levels $\{-\frac{15}{4}, -\frac{15}{4}, 0\}$ are identical in the two truncations, so the entire effect of the cutoff lives in the four cross-cell blocks. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \to -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

```
check:      the B=4 comparator on the same geometry gives
-1/12 with the reversed spectrum
```

References

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